Machine Learning for Predicting Trading Signals

1. Introduction

This project focuses on predicting **trading signals (Buy, Sell, Hold)** using machine learning techniques rather than forecasting stock prices directly.

Financial markets are highly volatile and difficult to predict. Instead of modeling raw prices, this project transforms price data into **technical indicators**, which are then used to generate trading decisions. Machine learning models are trained to learn patterns in these signals.

Objective

- Convert stock data into meaningful indicators
- Generate Buy/Sell/Hold signals
- Train models to predict these signals
- Evaluate model performance

2. Methodology

2.1 Data Preparation

The dataset consists of historical stock prices, specifically the **closing price**.
Steps performed:

- Converted Date column to datetime format
- Sorted data chronologically
- Removed missing values

Important:
Data was **not shuffled** to preserve time-series order.

2.2 Feature Engineering

Feature engineering is the most critical part of this project.

📊 **MACD (Moving Average Convergence Divergence)**

MACD captures **trend and momentum**.

Formula:

MACD = EMA(12) − EMA(26)

- EMA(12): short-term trend
- EMA(26): long-term trend

A **signal line** is calculated as:

Signal Line = EMA(9) of MACD

Trading Logic:

- MACD > Signal Line → Buy
- MACD < Signal Line → Sell

RSI (Relative Strength Index)

RSI measures **overbought and oversold conditions**.

Formula:

RSI = 100 − (100 / (1 + RS))

Where:

RS = Avg Gain / Avg Loss

Interpretation:

- RSI < 30 → Oversold → Buy
- RSI > 70 → Overbought → Sell

2.3 Signal Creation (Label Engineering)

Signals are created using both indicators:

MACD	RSI	Final Signal
Buy	Buy	Buy (1)
Sell	Sell	Sell (-1)
Mixed	Mixed	Hold (0)

Important Insight:
We are **creating our own labels**, meaning the model learns from rules we defined.

2.4 Machine Learning Problem

- **Features (X):** MACD, Signal Line, RSI
- **Target (y):** Final Signal

This is a **classification problem**, not regression.

2.5 Train-Test Split

- 80% training data
- 20% testing data
- No shuffling

Reason:
Time-series data must respect **past → future order** to avoid data leakage.

3. Models Used

3.1 Logistic Regression

- Simple and interpretable
- Works well for linear relationships
- Limited in capturing complex patterns

3.2 Random Forest

- Ensemble of decision trees
- Captures non-linear relationships
- More powerful and accurate
- Less interpretable

3.3 Support Vector Machine (SVM)

- Finds optimal decision boundaries
- Works well in complex spaces
- Computationally expensive

4. Model Evaluation

Models were evaluated using:

Accuracy

- Percentage of correct predictions
- Can be misleading if classes are imbalanced

Precision

- How many predicted signals are correct
- Important to avoid false trades

Recall

- Ability to detect actual signals
- Important to capture opportunities

F1 Score

- Balance between precision and recall

5. Results & Model Comparison

Model	Accuracy	F1 Score	Observation
Logistic Regression	0.74%	0.73	Simple but limited
Random Forest	0.73%	0.73	Best performance
SVM	0.74%	0.74	Good but slower

Analysis

- Random Forest performed best due to its ability to capture non-linear relationships
- Logistic Regression struggled with complex patterns
- SVM performed well but required tuning

6. Insights

Are signals predictable?

Partially.
Models can identify patterns, but predictions are **not always reliable** due to market randomness.

Which indicator is stronger?

- MACD captures **trend direction**
- RSI captures **extreme conditions**
- Combined signals are more reliable than individual ones

Key Observation

Combining indicators:

- Reduces false signals
- Improves decision quality

7. Key Concepts Discussed

Domain Knowledge Integration

Technical indicators help transform noisy price data into meaningful signals.

Label Creation (Critical)

- Labels are manually defined
- Model learns human-designed logic
- Bias in rules affects predictions

Time Series Caution

- No random shuffling

- Prevents future data leakage

Overfitting in Finance

- Models may perform well on historical data
- Fail in real-world trading

Interpretation of Signals

- Buy ≠ guaranteed profit
- Models assist—not replace—decision making

8. Limitations

Lagging Indicators

- MACD and RSI rely on past data
- Signals may come too late

Market Complexity

Stock prices are influenced by:

- News
- Economic conditions
- Global events

Class Imbalance

- Too many “Hold” signals
- Can bias model predictions

Missing Factors

Model does not include:

- Volume
- News sentiment
- Macroeconomic indicators

9. Critical Thinking Discussion

Are MACD and RSI enough?

No. They provide limited insight and should be combined with other indicators.

What happens during market crashes?

- Indicators fail due to extreme volatility
- Models become unreliable

Should “Hold” be removed?

- Removing it simplifies classification
- But reduces realism

Is accuracy the best metric?

No.
Precision and recall are more important in trading to:

- Avoid bad trades
- Capture good opportunities

10. Conclusion

This project demonstrates how machine learning can be applied to financial trading signals. While models can learn patterns from technical indicators, their predictive power is limited by market complexity.

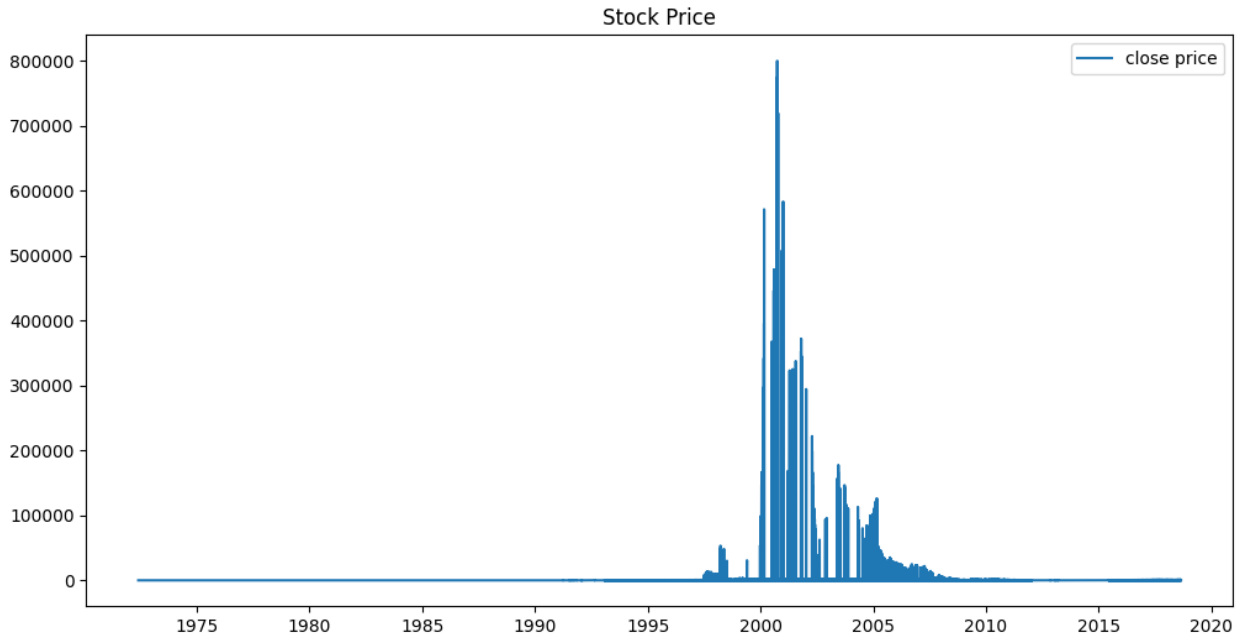
Final Takeaways:

- Feature engineering is more important than the model
- Combining indicators improves signal quality
- Machine learning assists decisions but does not guarantee profit
- Financial prediction remains inherently uncertain

11. Future Improvements

- Add more indicators (Bollinger Bands, Volume)
- Use deep learning models (LSTM)
- Perform backtesting for profit evaluation
- Incorporate real-world factors (news, sentiment)

GitHub Link

https://github.com/kadiwala1234/DAAI-Project-3.git

 # Visualization